DENDRITRON RECURRENT ENGRAM

Address Knowledge Sparsely.
Move Thought Geometrically.

A CPU/ARM-first recurrent architecture joining signed HarMax attention, expert-owned reasoning branches, composed LoRA skills, and frozen conditional memory.

PHRASE ENGRAMS

1,000,000 exact rows
layer08[2048] BF16
layer24[2048] BF16
separate 2g / 3g banks

DEFINITION BANK

1,532,746 sense rows
layer02[2048] BF16
31 immutable shards
canonical JTD frame

LIVE CORE

signed HarMax field
expert-owned branches
composed LoRA skills
effective depth 2R

Retrieve knowledge sparsely. Move the live state geometrically.
Reuse two physical blocks for recurrent CPU/ARM reasoning.

Technical architecture brief | Revision 4.1 | 16 August 2026

Dendritron Recurrent Engram synthesis and integration: Ryan Carson

01 | SYSTEM

Memory locates. Geometry moves. Recurrence reasons.

A FROZEN MEMORY PLANE AND AN ACTIVE REASONING PLANE MEET INSIDE EVERY RECURRENT VISIT

OFFLINE DONOR

Qwen phrase passes
Qwen definition readouts
token-embedding export
GPU cost paid once

FROZEN MEMORY

2g / 3g phrase Engrams
layer-2 definition senses
JTD projection assets
immutable manifests

LIVE REASONING

surface + LNGram routes
signed HarMax movement
composed LoRA skills
two reused blocks

Dendritron gives memory and reasoning equally explicit contracts. Frozen phrase vectors and definition-sense landmarks supply bounded coordinates. The evolving 2,048D live state carries the current problem, candidate relations, branch evidence, and intermediate conclusions through repeated visits to two physical blocks.

MEMORY PLANE

Exact suffix lookup, Hash-Engram miss coverage, LNGram definition addressing, and immutable donor rows. Retrieved values enter the live state through calibrated gated residuals.

REASONING PLANE

Geometric attention creates signed attraction and repulsion. Expert-owned branches propose and test relations; composed skills move the state through two DeepLoop blocks.

ONE VISIT

live state -> JTD query -> LNGram and surface gather -> HarMax probability field -> signed joint-space movement -> expansion or contraction block -> composed skill -> bounded expert soma -> next live state

02 | BUILT ASSETS

Built memory and alignment assets

PUNCTUATION-V2 PHRASE BANKS, DEFINITION LANDMARKS, TOKEN SEEDS, AND REAL CPU JTD FITS

TRIGRAM ENGRAM

500,000 exact 3-word rows
layer08[2048] + layer24[2048]
independent row and shard space

BIGRAM ENGRAM

500,000 exact 2-word rows
layer08[2048] + layer24[2048]
independent row and shard space

Asset	Measured build evidence	State
Punctuation-v2 inventory	421,332 bigrams and 388,443 trigrams retained; 78,668 and 111,557 replaced	COMPLETE
Corrected phrase bank	809,775 paired rows copied + 190,225 freshly extracted = 1,000,000 rows	COMPLETE
Qwen symbol seeds	248,077 x 2,048 BF16 input-embedding rows	COMPLETE
Definition landmarks	1,532,746 x 2,048 BF16 layer-2 sense rows; 31 shards; about 5.8 GiB	REPORTED
JTD anchor triples	32,768 bigrams + 32,768 trigrams; layer-2 reference with layer-8/24 sources	COMPLETE
Real CPU JTD fit	J8 loss 21.849 to 0.372; J24 loss 37.923 to 0.844	COMPLETE

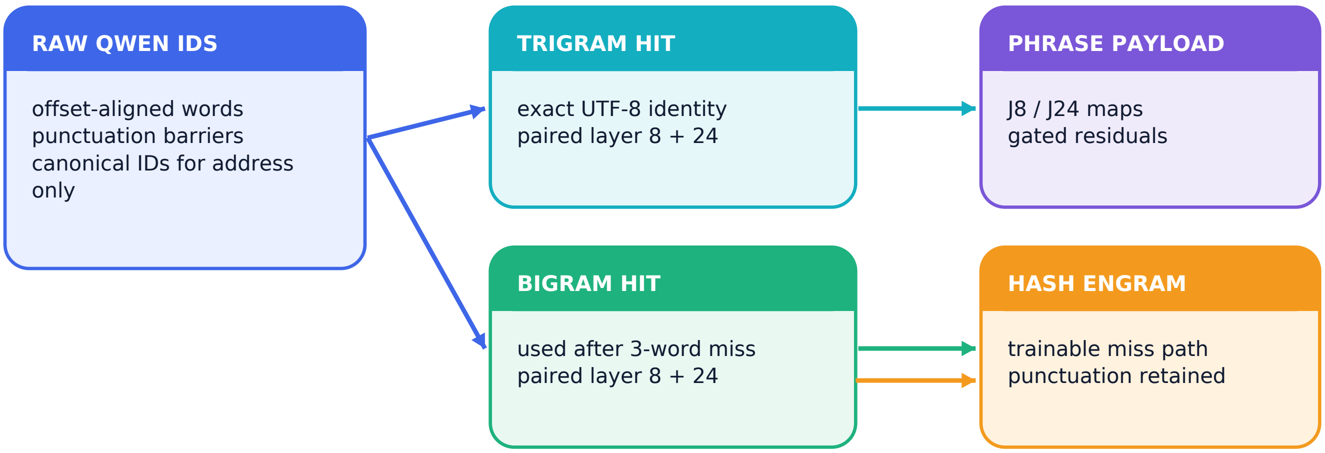
ASSET LIFECYCLE

Canonical commits reference frozen assets by manifest identity, revision, shape, dtype, and cryptographic hash. New Engram, LoRA, router, expert, and provenance deltas remain compact and versioned.

03 | SURFACE MEMORY

Exact phrases and constituent senses

LONGEST EXACT 3-TO-2 PHRASE LOOKUP COEXISTS WITH PER-WORD DEFINITION SEEDING



CONSTITUENT SENSE SEEDING

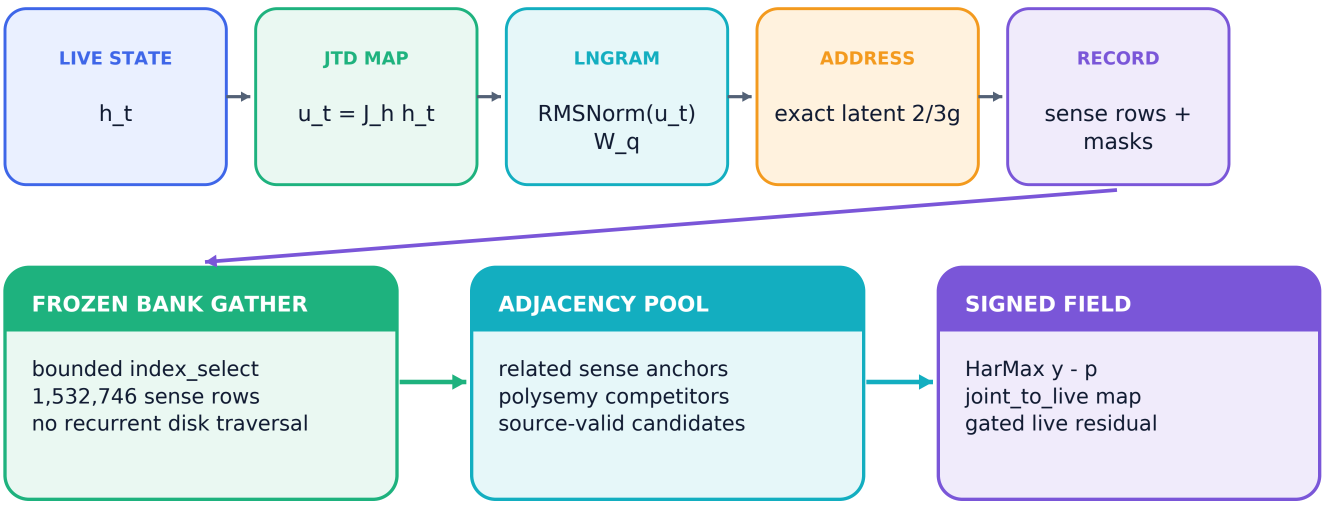
Every complete input or decoded-reasoning word emits its dictionary sense IDs into the bounded definition pool, even when a longer exact phrase supplies the primary phrase-memory row. Sense IDs are deduplicated while source, span, and word-order provenance remain attached.

Layer	Address	Payload	Role
Phrase	Exact 3-word, else exact 2-word	Frozen layer-8 and layer-24 vectors	Narrow contextual memory
Constituent	Every complete word	All retained definition-sense handles	Semantic neighborhood seeds
Miss	Canonical local Qwen-ID suffix	Trainable Hash-Engram rows	Coverage outside frozen phrase bank

04 | DEFINITION MEMORY

JTD-aligned LNGram definition memory

CONTINUOUS ALIGNMENT, DISCRETE SPARSE SELECTION, THEN CONTINUOUS EVIDENCE MOVEMENT



IMPLEMENTED / REPORTED

Real J8/J24 CPU fits, live and definition-frame maps, complete bank attachment, and safe gather from supplied sense-row tensors.

NEXT INTEGRATION

O-019 route objective, populated address records, O-021 target evidence, and replacement of centroid attraction with the full signed field.

The latent address integer and dictionary row integers remain separate namespaces. JTD places the evolving live query in the frozen layer-2 definition frame before LNGram forms its address. The retrieved vectors already occupy that frame, so the bridge performs selection and gathering rather than a second semantic map.

RESEARCH BASIS AND DENDRITRON EXTENSION

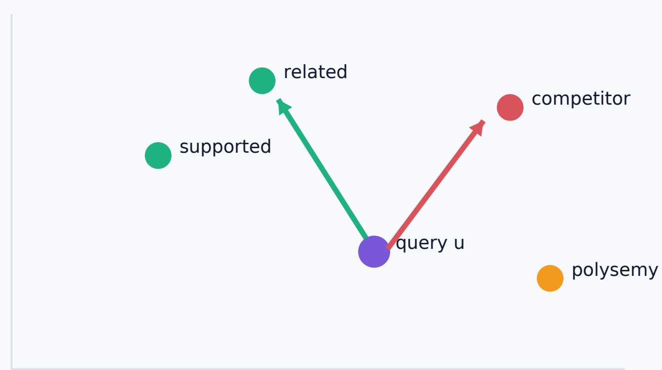
LNGram contributes tokenizer-independent discrete routing from continuous hidden states. Locality Preserving Joint Transfer contributes source-specific continuous alignment. Dendritron binds them through bounded definition-sense handles and keeps phrase Engrams and Hash-Engram as independent memory branches.

05 | GEOMETRIC ATTENTION

HarMax turns evidence into a force field

EUCLIDEAN PROXIMITY, HARMONIC RESIDUAL, AND ITS NEGATIVE DERIVATIVE CONTROL LIVE MOVEMENT

JOINT-SPACE CONTRACTION POOL



SIGNED FIELD

$D_q = ||u - c_q||_2^2 + \mathrm{lambda}_p\ \mathrm{delta}_q^2 + \mathrm{eps}^2$

$p_q = D_q^{-n/2} / \mathrm{sum}_j\ D_j^{-n/2}$

y_q = normalized positive supported evidence

$\mathrm{rho} = -\mathrm{sum}_q\ y_q \log p_q$

$\mathrm{Delta}\ u = n\ \mathrm{sum}_q\ (y_q - p_q)\ (c_q - u) / D_q$

h <- h + gate * joint_to_live(Delta u)

y > p

Underrepresented supported evidence attracts the live query.

y < p

Unsupported or excess proximity produces repulsion.

y = 0

Competitors remain in the denominator and supply contrast.

HARMONIC CONTROL CONTRACT

Uniform scaling of the completed distance vector cancels inside HarMax; causal and epsilon terms share the same normalized construction. Harmonic Loss supplies the finite-center and interpretability motivation. Dendritron locks the recurrent probability mass, residual, and signed derivative above; O-021 supplies definition-pool target evidence.

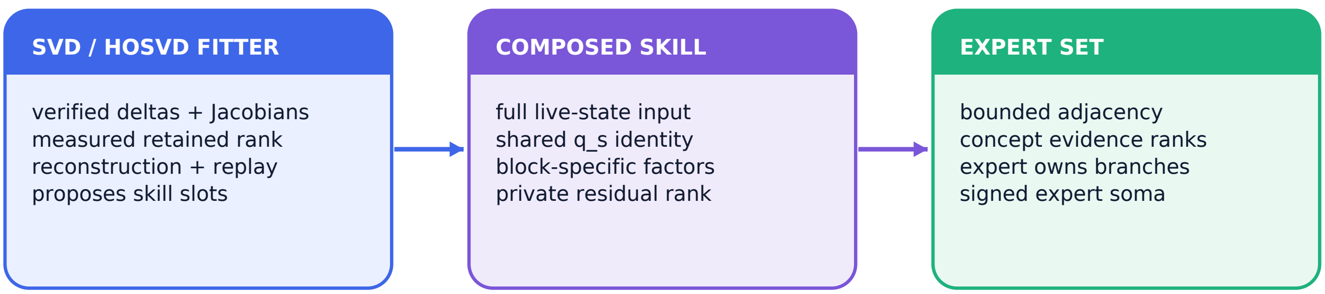
06 | PROCEDURAL COMPILER

Shared modes, private residuals, bounded experts

VERIFIED TRAJECTORIES COMPILE REUSABLE SKILL COORDINATES WHILE PRESERVING SPECIALIZED UPDATE ENERGY

COMPLETE SKILL LORA FOR SKILL s AND BLOCK b

$$\text{Delta } W_{s,b} = B_b^{sh} \, \mathrm{diag}(q_s) \, A_b^{sh} + B_{s,b}^{priv} A_{s,b}^{priv}$$



Element	Contract
Shared term	Block-specific A/B factors; dense q_s spans the full shared basis and preserves skill identity across blocks.
Private term	x = (I - UU^T) delta stays with the selected skill's private factors and verified experience.
Compiler	SVD or HOSVD is evidence-selected; retained modes follow energy, reconstruction, task success, and replay preservation.
Capacity + route	Up to 32 initial skill slots; populated masks control use. skill_to_experts defines the bounded expert set.

CURRENT TREE: 176 TESTS REPORTED

07 | CONTINUAL LEARNING

Frozen runtime, verified offline updates

SUCCESSFUL TRAJECTORIES BECOME PROVENANCE-BEARING EXPERIENCE AND CAREFULLY VALIDATED PARAMETER DELTAS

SUCCESS
task outcome + evidence
→
CAPTURE
detached Delta h + IDs
→
FIT
private skill buffer
→
VERIFY
replay + regression
→
COMMIT
versioned delta registry

LIVE EXECUTION

Core weights, shared basis, skill coefficients, private factors, experts, and router anchors are frozen during inference and task execution.

OFFLINE WORKING BUFFER

Only verified success authorizes a private-skill update. Replay, regression, provenance, and version checks precede commit.

CANONICAL ASSET-REFERENCED CHECKPOINT

Stores compact changing state

- new or updated Engram records
- composed shared/private LoRA factors and q coefficients
- router, expert, and branch registries
- success evidence, verification, and provenance

References immutable assets once

- token-embedding matrix
- definition bank
- layer-8/layer-24 phrase banks
- fitted JTD projections
- identity, revision, shape, dtype, and hash

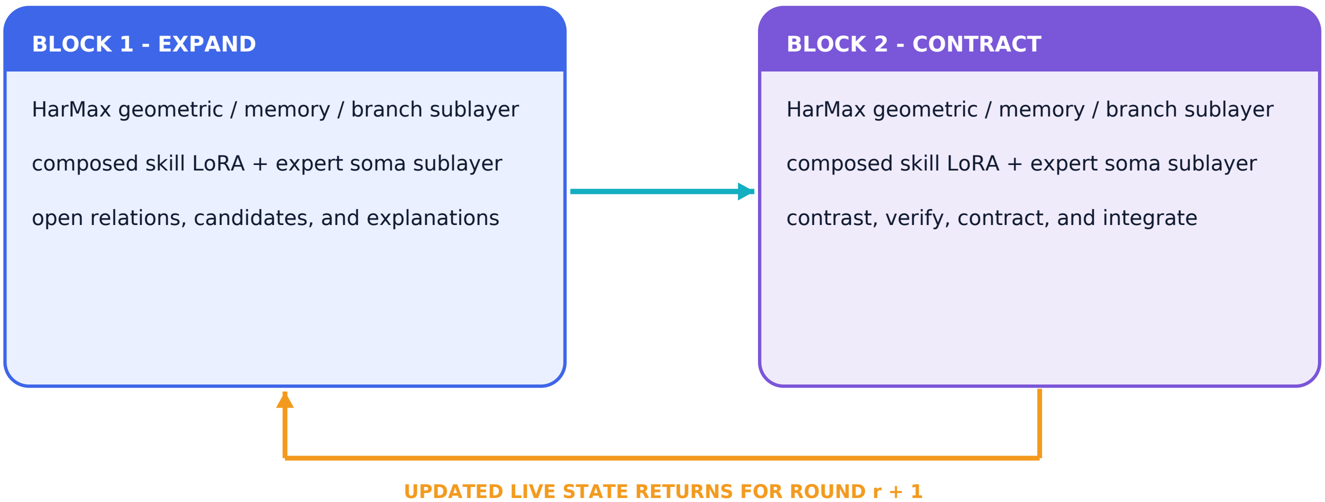
CONTENT AND PROCEDURE STAY SEPARATE

Personal facts and successful outcome traces belong in experience Engrams. Shared and private LoRA factors carry procedural skill. Frozen lexical and phrase assets retain byte identity across runtime and offline fitting.

08 | REASONING ENGINE

Expansion, contraction, and typed branches

EVERY ROUND PROPOSES RELATIONS, TESTS THEM GEOMETRICALLY, AND RETURNS AN AUDITABLE LIVE-STATE UPDATE



DEDUCT
premises to conclusion

CAUSE
cause, mechanism, effect

CONTRAST
separate candidates

COUNTER
change one condition

ABDUCE
rank explanations

ANALOGY
transfer relations

DEEPLOOP STABILITY CONTRACT

$K = 2$, $N = KR = 2R$, and two residual sublayers give $M = 2KR = 4R$. Every visit uses Post-RMSNorm with $\alpha = \sqrt{2N}$; designated residual matrices use $\beta = 1 / \sqrt{8N}$. The measured contract is $M \kappa_R (\beta / \alpha)^2 = O(1)$.

EXPERT BRANCH CONTRACT

Each selected expert instantiates typed branches from the live state, memory anchors, and active skills. Every branch returns derivative movement, harmonic residual, signed evidence, and an exact trace; signed L1-normalized branch movements form the expert soma.

09 | EXECUTION CONTRACT

CPU/ARM execution and the 25/75 law

THE LIVE PATH FETCHES BOUNDED MEMORY AND EXECUTES BOUNDED CONDITIONAL COMPUTATION

FIXED SPARSE-CAPACITY LEDGER

25% MEMORY	75% CONDITIONAL COMPUTE

OFFLINE GPU

Qwen phrase passes, definition readouts, and token-embedding export.

LIVE CPU / ARM

JTD, LNGram, sparse gathers, HarMax, LoRAs, experts, blocks, and decoding.

BOUNDED MEMORY

Attach frozen banks once; gather selected rows per visit; keep SQLite/shards outside the unroll.

SHARED METRIC

Ordinary Euclidean distance controls HarMax, concept scoring, route evaluation, and vocabulary ranking.

Runtime choice	Live effect
Exact address and bounded gather	Memory-bank size does not expand the active candidate pool.
Two reused blocks	Reasoning depth grows through visits rather than stored layers.
Low-rank skills + sparse experts	Only the routed procedural neighborhood executes.
Softmax-free operators	Routing, memory fusion, expert selection, and tied Euclidean decoding share a CPU-friendly contract.

10 | HONEST BOUNDARY

Built boundary and next integration

COMPLETED ASSETS AND VERIFIED MODULES STAY DISTINCT FROM THE REMAINING DEFINITION-ROUTE ASSEMBLY

Punctuation-v2 phrase inventories and 1M paired donor rows	COMPLETE
Token seeds, JTD anchor export, and real CPU J8/J24 fits	COMPLETE
1,532,746-row definition bank and supplied-sense-row attachment	REPORTED
Two-block HarMax core and tied Euclidean output	CPU VERIFIED
Composed LoRA / expert / provenance current tree	176 TESTS REPORTED
LNGram address to populated bounded sense-row records	NEXT: O-019
Definition-pool target evidence and full signed field	NEXT: O-021
Typed branch executors, language training, and ARM benchmarks	MILESTONE

IMMEDIATE BUILD ORDER

1. Select and verify O-019 routing supervision. 2. Populate bounded address records and measure route stability. 3. Lock O-021 evidence construction and connect the signed definition field. 4. Prove end-to-end constituent-sense payload materialization. 5. Train, benchmark, and optimize the typed branch system on ARM.

EVIDENCE LANGUAGE

COMPLETE records a finished artifact. CPU VERIFIED records an executed reference module. REPORTED preserves evidence supplied by the active build or coding-agent worktree. CURRENT INTEGRATION names the live assembly boundary. MILESTONE names an engineering or experimental target. The following page maps every supplied research source to its precise architectural, training, evaluation, or scientific role.

11 | SOURCE TRACEABILITY

Where each research source enters

RUNTIME FOUNDATIONS ARE SEPARATED FROM TRAINING, EVALUATION, GEOMETRY, AND SCIENTIFIC MOTIVATION

Source	Dendritron use	Class
Harmonic Loss Trains Interpretable AI Models — 2502.01628	HarMax mass, harmonic residual, signed derivative field; finite-center results motivate evaluation.	RUNTIME
DeepLoop: Depth Scaling for Looped Transformers — 2607.13491	Two tied blocks, repeated visits, kappa_R diagnostic, and conservative p = 1/2 scaling.	RUNTIME
Conditional Memory via Scalable Lookup — 2601.07372	Canonical token projection, deterministic conditional memory, bounded sparse lookup.	RUNTIME
Memory Grafting — 2605.20948	Offline donor states, exact longest-suffix phrase memory, projection/gating, hash fallback.	RUNTIME
Lngram — 2605.24869	Hidden-state discretization and exact latent 2/3-gram addresses.	RUNTIME
Locality Preserving Joint Transfer — 1906.07441	Source-specific maps into a shared locality-preserving semantic frame.	RUNTIME
Shared LoRA Subspaces — 2602.06043	Shared procedural factors, coefficients, private residuals, and consolidation.	RUNTIME
User as Engram — 2606.19172	Separation of sparse content memory from shared reasoning skill.	DESIGN
Collision-Free Hot-Tier Extension — 2601.16531	Route-stratified collision and gate-credit diagnostics for Engram evaluation.	EVAL
Bilevel Optimization for NAS — 2606.29582	Offline selection framework for fitter, router, and consolidation hyperparameters.	TRAINING
Similarity-Guided Curriculum — 2607.11591	Similarity-banded offline curricula and interface-preservation checks.	TRAINING
A*-Inspired Batch Selection — 2607.15745	Priority and reuse-penalty ideas for verified trajectory and replay scheduling.	TRAINING
Distinct Tasks Engage a Shared Neural Subspace — doi:10.64898/2026.04.24.720703	Empirical motivation for a stable shared procedural subspace across tasks.	SCIENCE
A Shared Valence Axis Across LLMs and EEG — 2606.00129	Shared-axis and saturation diagnostics for alignment and consolidation studies.	SCIENCE
From Directions to Cones — 2505.21800	Multidimensional semantic-cone hypotheses, causal probes, and anchor-neighborhood evaluation.	GEOMETRY
Geometric Unification with Concept Cones — 2512.07355	Cone containment and alignment metrics for supervised and discovered directions.	GEOMETRY
Extinction Depth and q-ary Error-Correcting Codes — 2607.19566	Finite-horizon inspiration for future route-collision certification.	FUTURE